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RESEARCH-ARTICLE

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A Pilot Study of User Preferences of Posture and Display Technologies in Virtual Reality Exercise Games

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ABSTRACT

With the continuous development of extended reality (XR), encompassing augmented reality (AR), virtual reality (VR), and mixed reality (MR), the increasing application of VR, especially combined with game technology in the health area, is trending. Due to this development, academia and industry have rising research and practices focusing on VR exercise game applications and their evaluation. This paper presented a pilot study addressing the comparison of user preferences in using VR exercise games. Eight volunteering participants with VR or rowing experience were involved in the pilot study. Their responses on using different postures (standing or sitting), display devices (VR or a large screen), and game tasks (collect coins vs distance travelled) were explored, as well as feedback suggestions for the study and VR games. The pilot study revealed the opportunities and challenges to enhance the VR exercise games, user experience, and performance. It tested the feasibility and duration of each session and potential improvements that could be made for the main experiment, including the instructions, game environments, process, devices, data gathering, and analysis methods.

CCS CONCEPTS

• **Human-centered computing** → **Usability testing; Laboratory experiments; User studies; Displays and images.**

KEYWORDS

virtual reality, exercise game, user preference, posture, head-mounted display, large screen

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1 INTRODUCTION

In recent years, people's pursuit of health has driven the development of exercise games. An exercise game is a game that requires

the user to participate in physical activity [19]. Some are developed for recreational purposes but have health-promoting effects, such as Pokémon Go [19]; some are designed with health goals, such as avoiding a sedentary lifestyle [19]. Virtual reality (VR) exercise games are based on VR technology, a significant application field of immersive technology increasing in recent years [4]. As evidenced by an industry survey in extended reality (XR), the report pointed out that the two main themes in the application field are games and health [16]. Academic research also pays attention to VR exercise games. In the article by Yu et al. [5], it is pointed out that there is evidence that VR exercise games benefit both physical and mental health. Thus, VR exercise games are often used in studies to prevent and treat various diseases, including childhood obesity prevention [8], type 2 diabetes prevention [15], and post-stroke rehabilitation [18].

While increasing designs, developments, and applications of VR exercise games, user evaluation and testing have also attracted much attention. As Yu et al.'s review pointed out, over 73% (65/88) of research includes an evaluation section, in which the most used evaluation method is questionnaires to obtain subjective feedback. Their study also mentioned that the test standards in such papers often correspond to software quality indicators, such as usefulness, usability, and user experience [4]. Among these, a few user preferences were tested in such studies, e.g., display device [20] and using postures [7]. In addition, surveys on immersive technologies' game applications for the general public and healthcare-related personnel could also find more care for user experience and devices. The most significant obstacle of immersive techniques to mass adoption was the user experience, according to over 83% of participants [3]. The main problems came from the device, such as physical discomfort, heaviness, and space requirements [3].

1.1 Aim and research questions

This study has the scope of VR exercise games and attempts to use a combination of experiments and questionnaires to explore users' preferences for such software regarding exercise postures, displays, and the game task performance within them. At the same time, the study pre-tests and contributes to a larger-scale experiment on the feasibility evaluation of the experimental object, environment, approach, device, and analysis method and collects suggestions for improvement of the process and setup. Researchers could become familiar with the study process and reduce unnecessary effort from participants and researchers while reducing the dissipation of study resources in the main study [17]. To achieve the above research aims, there are two objectives: 1) testing the users' preferences on exercise postures, displays, and game task performance, and 2) obtaining the comments of the study and games by open questions. Our approach

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is to explore the opportunities and challenges of VR exercise games with user experience and usability. The research objectives are aligned with the following research questions: RQ1) What are the users' preferences for the exercise postures (sitting vs standing) in VR exercise games? RQ2) What are the users' preferences for display devices (HMD vs large screen) in VR exercise games? RQ3) What are the users' VR exercise game tasks (collecting coins vs distance travelled) performance in different postures or display devices?

1.2 Background and related work

In a previous XR in health games literature review, Yu et al. pointed out that the difference between immersive VR (devices using HMD or CAVE) and non-immersive VR (running on a large screen, laptop, computer, tablet, mobile or TV) is whether it cuts off the perception of the real world to some extent [4]. They found a few articles [18, 20] that compared the VR display device preferences and obtained similar results between papers. Each device had its advantages and disadvantages. HMD VR could provide a more immersive experience. However, the simulator sickness problem cannot be ignored. However, some non-immersive VR had a lower immersive experience but higher accessibility and portability [4]. The device's choice should depend on the target user and application purposes. As strong evidence, we found a research team changed the VR display device from a CAVE to a large screen considering the home-based exploitation (using a normal screen, a less-expensive technology, instead of a CAVE and promoting the participation of the elderly) [1].

In the above review [4], Yu et al. also listed the twelve quality indicators most involved in the included papers. The three most frequently measured indicators include usefulness (30), usability (14), user experience (14), and sickness and discomfort (5). However, using scales designed by researchers outnumbered user feedback collection using standardized scales or questionnaires based on variants of standardized scales. However, among the statistical quality indicators, there were still some standardized scales used by more than half, such as the System Usability Scale Questionnaire (SUS) (9/14) [2] and Simulator Sickness Questionnaire (SSQ) (3/5) [9].

As mentioned above, there were a few studies that paid attention to user preferences. Some papers compared the preferences of immersive and non-immersive VR displays (laptop/TV/HMD/large screen) [18, 20], and their results were similar. Each display had its pros and cons [4]. The advantage of a laptop is that it is more common and more accessible to carry; however, the size of its screen limits the display of objects, and it cannot follow head movements to change the perspective [20]. Large screens and TVs can provide wider viewing angles, but they are still not as immersive as HMDs [20]. Yoo and Kay tested the flexibility and availability of HMD. They mentioned some of its problems, such as user injury risk and the fit problem of HMD [20].

The study by Hu et al. compared the impact of standing and seated viewing of 360° videos on subjective quality assessment when using an HMD. They noted that the average rating time consumed for quality assessment was similar in different postures. A higher resolution was obtained between different quality levels

when viewing from a seated position, and simulator sickness was significantly reduced when seated [7].

1.3 Overview

The rest of the paper is organised as follows. Section 2 describes the experiment equipment, software, environment, and procedures for this study. The participants and open questions used for feedback collection are also described in this section. Section 3 presents the results, including experimental and open questions results. It is designed to answer the research questions of the pilot study. Section 4 discusses the results and compares them with related works. Finally, the conclusions and future work are presented in Section 5.

2 METHODS

This pilot study is a pre-study to test the feasibility of the proposed method and process to avoid potential problems when conducting the main experiment [17]. There are several aspects tested: 1) process (evaluating the feasibility of the steps), 2) resources (assessing time cost and devices requirements), 3) management (focusing on data management and training researcher), 4) scientific (testing the research and analysis methods).

Ethical approval has been obtained for this study. The following section illustrates the experiment procedure, participants, used platform and VR games, and devices and equipment. Except with the questionnaires mentioned before (SUS, SSQ), to obtain more user experience, this study involved the game experience questionnaire (GEQ) and the NASA Task Load Index (TLX).

2.1 Procedure

The experiment had three main parts. Except for the steps of invitation and scheduling, all others were conducted in a lab. In the first part, participants were introduced to the experiment process and tasks before they could sign the informed consent. Then, they were asked to perform a simple visual acuity test, including short-sightedness and colour blindness, to ensure vision problems were not a factor in the experiment. The large-screen version of the VR games used in this experiment was applied in the training step. The researcher demonstrated how to play the games (including controlling the boat to move forward, turn, retreat, obtain coins, and escape after hitting the riverside). Participants could play the standing and sitting rowing games to familiarise themselves with the controls. After the training, a short demographic questionnaire and SSQ aimed to collect data such as age group, gender, and experience of VR games and rowing activity, as well as to ensure the participants were not discomfort and could continue the experiment.

The second part is the experiment. The full experiment process includes four sessions to evaluate the user preferences of different display devices and postures, Session 1 (S1): Standing HMD, Session 2 (S2): Standing large screen, Session 3 (S3): Sitting HMD, and Session 4 (S4): Sitting large screen. These sessions would be conducted in different orders facing the participants to reduce order bias. Each session began with the device setup and ended with the questionnaires, covering three minutes of playing the games and around eight minutes of completing the questionnaires. Each session was followed by a questionnaire composed of the SUS, GEQ, TLX, and

SSQ. After the final session, three additional questions (7-Likert scale) were also asked to obtain user feedback on the overall feel of using the different devices and postures and ranking the need for sound and background music.

Besides the quantitative data collected from the experiment, we also recorded some qualitative data from the observation of participants. Observation as a research method could usually be structured and unstructured. We use unstructured observation in our study to observe and take notes of participants' behaviour during their experience [14].

Finally, the HMD would be disinfected between each experiment and participant to minimize the spread of any potential illnesses. This was done using the Cleanbox Technology. The room was also ventilated between each experiment. To collect comments, after the experiment, participants were asked to point out suggestions for improving the experiment and the VR games.

2.2 Participants

Eight volunteering participants were invited to the pilot study. All of them were academic research students. They were aged 18 to 45, with an equal number of females and males. On the Likert-scale rating system of 0 (no experience) to 5 (rich experience), four participants rated their rowing experience and seven rated VR experiences as less than or equal to 2. Only one participant considered he/she has extensive VR experience (score 5) or boating experience (score 4).

2.3 Experiment setup

The experimental platform (iMotions Lab) was used in the study for the questionnaire and its data collection from iMotion A/S, Copenhagen, Denmark. The computer configuration referred to the minimum requirements and recommended setup of the desktop computer provided by iMotions, combined with the running conditions of the HMD (HTC VIVE Pro eye) used in the experiment, the computer configuration used in the experiment. The large-screen version test used SAMSUNG Flip Pro 65" WM65B. Moreover, the Cleanbox CX1 was also used in this study for HMD disinfection by UVC LED.

Unity 2018.4.11c1 (64-bit) was used to construct the VR rowing games in this pilot study. The interfaces and navigations are shown in Figure 1. The game's design and development referred to the research team's previous prototyping article [6] and realized the basic functions of boating for physical exercise. The controllers are banded on a wooden stick to simulate the boating paddle to interact with the system (as shown in Figure ??).

The games consist of two modes, corresponding to two rowing postures. The two usage postures were tested for the experimental purpose (stand-up paddle board: standing, kayaking: sitting). Each game mode contains two maps (beginner and challenger), which correspond to different numbers of the river turns to differentiate the difficulty of the games. In this experiment, only the beginner mode was used. Players row the stand-up paddle board or the kayak in the river from the first-person perspective. Different numbers of gold coins are scattered in the middle of the river, and the player can control the boat's direction to pass through the gold coins to get them. In addition to simulating the environment, the game interface

provides a timer, gold coin accumulation record, driving distance record, and buttons for game pause, exit, and return to the main interface. The records for the gold coin accumulation and driving distances for each second were recorded as in-game data.

The main game mechanics of experiment games are based on physical activities and game tasks, and they consist of modes and roles, rewards and punishments, time management, and disadvantage compensation. The game can be run on the HMD or the large screen to test user preferences for different display devices.

3 RESULTS

This section shows the experiment and open questions results. All numerical data was processed separately by descriptive statistics and hypothesis testing to address the RQ1, RQ2, and RQ3.

After gathering the experimental data, the scale scores were used to calculate the total score of each scale according to their calculation formula (GEQ and SSQ scores also include their respective sub-item scores). The in-game data (only use the final gold coin and distance travelled), overall scale data (whole feeling for HMD/large screen/sitting/standing in VR rowing exercise games), and the above total scores would be processed in descriptive statistics and hypotheses test.

To measure the user preferences for using different postures, we compared the data between S1/S3 and S2/S4, in the same way, fixing one variable and comparing the effect of the other variable. The device user preference evaluation compared the data between S1/S2 and S3/S4.

Descriptive statistics included normal distribution detection and calculation of mean and standard deviation. The normal distribution was used to determine whether it was a parametric test, thereby determining the test method. The mean and standard deviation were used to compare the user preferences. Neither the Kurtosis or Skewness values of scale scores, in-game data and overall scale data conformed to the normal distribution. Therefore, their hypothesis test should select methods in non-parametric testing. Because the research goal was to compare different choices of the same object (device: HMD/large screen; posture: sitting/standing; game task: coins/ distance), that is a single factor with two treatments. Furthermore, these result data were to be compared pairwise. According to Marascuilo and Serlin [12] and Montgomery [13], the characters of the results (one factor, two treatments, paired comparing) in this study matched the Wilcoxon test to perform hypothesis testing.

3.1 Score analysis of scales

Based on the description of RQ1, RQ2, and RQ3, we define the following hypotheses for scale scores. H_0 : There is no significant difference in the score of SUS/TLX/GEQ/SSQ results when changing the usage posture/display devices/game tasks. H_1 : There is a significant difference in the score of SUS/TLX/GEQ/SSQ results when changing the usage posture/display devices/game tasks. Calculate the rank sum d_i^+ s of all positive deviations (as T^+) and the rank sum d_i^- s of all negative deviations (as T^-) was the first step. Based on T^+ , T^- , and the number of paired samples n , H_0 could be determined whether H_0 rejected through the Wilcoxon signed rank test critical. If the lower value between T^+ and T^- was equal or under the corresponding value in the table, then H_0 was rejected.

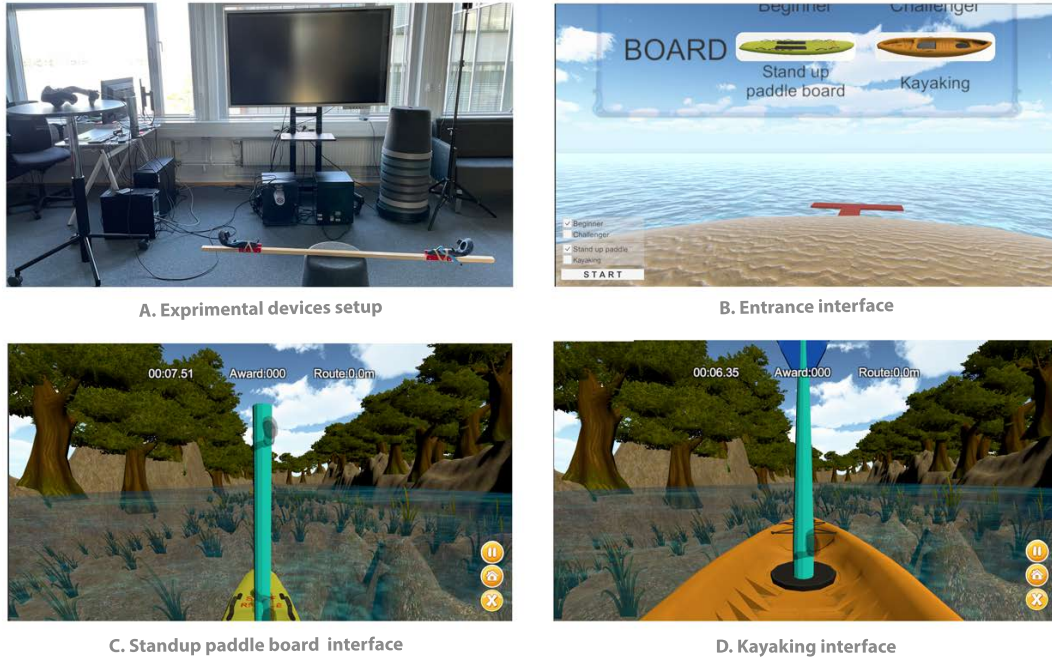


Figure 1: The screenshot of the VR rowing games.

Since the smallest in T^+ and T^- is 5, the test used a two-tailed test, $\alpha = 0.10$. The hypothesis test result is shown in Table 1.

As seen in Table 1, changing postures or display devices had no significant difference in the SUS score. However, the results in the TLX, GT and ST were different. HMD as a display did not change the GT scores, no matter in which postures. However, large screen/sitting/standing as the dependent variable changed the GT scores. Vertical comparison on the sub-items of GEQ: regardless of sitting or standing, when the display was fixed, there was no significant difference in the scores of Co, Fl, and Po, while there was a significant difference in the scores of Ch in both sports postures (sitting/standing). There were similar results in display devices compared: Te/Ne scores had no significant difference, while there was a significant difference in the scores of Co.

Furthermore, the standing posture led to significant differences in Se/Te/Po scores on different monitors, and this phenomenon appeared in the Fl indicator of sitting posture. In other words, when the posture remains unchanged in sitting, the scores of changing HMD and the large screen had significant differences in the Fl item. The Se scores had significant differences only when the large screen was used as the display. On the other hand, when using HMD, the score of Te/Ne had significant differences.

For comparing sitting and standing postures, there were no significant differences between HMD and large screen in all SSQ sub-items and total scores. The sub-item Na of SSQ was consistent with the total score ST in comparing HMD and large screen. Under the unchanged postures, the scores of different displays showed significant differences. For comparing devices, the scores of the Oc and Di indicators were on the contrary in a sense. Oc has no significant difference in the scores of the two devices, only when sitting was

the usage posture, while Di has a significant difference only in such cases. The TLX scores only changed on different postures but not on displays.

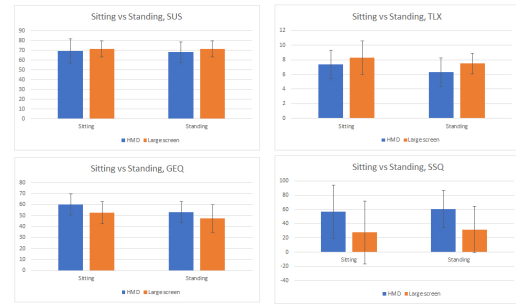


Figure 2: The mean and standard deviation of postures' comparing.

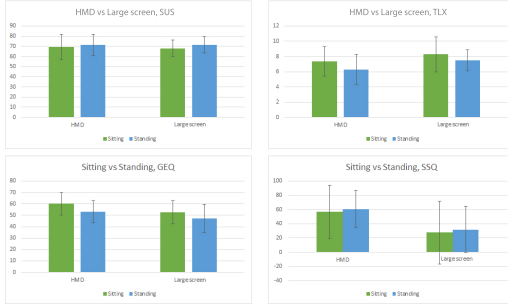
Combining Figure 2 and Figure 3, the scores of significant differences on SUS/TLX/GEQ/SSQ could obtain user preferences of usage postures/display devices and answer RQ1 and RQ2.

The SUS scores were not affected by posture and display. Different postures would affect the TLX score significantly, but not in the display devices. Whether using an HMD or a large screen, the users preferred the standing posture. There was no significant difference in GEQ scores in sitting and standing when using HMD. When using a large screen, the sitting posture was preferred over the standing one. Comparing the HMD and the large screen, regardless of the sitting/standing posture, users liked HMD more.

Table 1: The hypothesis testing results of scale scores

	SUS	TLX	GEQ								SSQ			
			Co	Se	Fi	Te	Ch	Ne	Po	GT	Na	Oc	Di	ST
Sitting HMD vs Standing HMD	H0	H1	H0	H0	H0	H1	H1	H1	H0	H0	H0	H0	H0	
Sitting Large screen vs Standing Large screen	H0	H1	H0	H1	H0	H0	H1	H0	H0	H1	H0	H0	H0	
Sitting HMD vs Sitting Large screen	H0	H0	H1	H0	H1	H0	H0	H0	H0	H1	H1	H0	H1	
Standing HMD vs Standing Large screen	H0	H0	H1	H1	H0	H0	H1	H0	H1	H1	H1	H1	H0	

Note: Co–Competence, Se–Sensory and Imaginative Immersion, Fi–Flow, Te–Tension/Annoyance, Ch–Challenge, Ne–Negative affect, Po–Positive affect, GT–GEQ total; Na–Nausea, Oc–Oculomotor, Di–Disorientation, ST–SSQ total.

**Figure 3: The mean and standard deviation of devices' comparing.****Table 2: The hypothesis testing results of in-game data**

		Coins	Route (m)
Standing	HMD vs Large screen	H1	H1
Sitting	HMD vs Large screen	H0	H1
HMD	Standing vs Sitting	H0	H0
Large screen	Standing vs Sitting	H0	H0

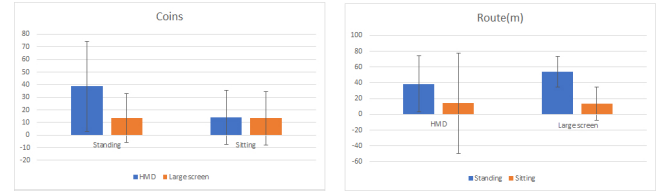
Furthermore, the SSQ scores were not changed significantly by different postures with the same devices. However, when using sitting/standing postures, SSQ scores were lower on large screen, that is to say, large screen's discomforts were less and users seemed to like it more.

3.2 In-game data analysis

The in-game data was processed in the same steps as scale scores. We compared the final coin numbers and boating distance in the standing/sitting postures and HMD/large screen displays. According to the kurtosis and skewness values in the descriptive statistics, the data do not fit a normal distribution. Thus, the Wilcoxon test was also the hypothesis test method for the in-game data. Since all the smallest in T^+ and T^- was 7, the test used a two-tailed test, $\alpha = 0.05$. The hypothesis test result is shown in Table 2.

When playing VR sports games in a standing or sitting position, the display (HMD and large screen) will affect the completion of game tasks, while when the display is fixed, the usage posture has no impact on the game strategy. Combined with figures in 4, regardless of whether the user was standing or sitting, the number

of coins obtained and the rowing distance value are higher when using the HMD.

**Figure 4: The mean and standard deviation of coins and travel distance.**

3.3 Overall scale data analysis

Following the same data processing process, the hypothesis test results of the overall scale data show no significant difference in users' feelings about HMD/large screen and sitting/standing. The average value of HMD and large screen score was equal, and standing (4) was higher than sitting (3.75).

3.4 Feedback analysis

The feedback comments came from two channels (observed in the experiment) and could be grouped into two categories (suggestions for the study and VR games). There were 11 and 18 users' suggestions for the experiment and VR games, while 21 and 5 comments for the experiment process or analysis were obtained from the researchers' perspective. By gathering and summarising all feedback, we grouped them and ranked priority (1 to 3, priority from high to low) for the following work. The comments mentioned over one time were game operation problems (12), optimizing the game (8), and simplifying the questionnaire structure (7). clarified the experimental procedures/time required/materials (5), the paddle problems (5), preparing notes and examples for questionnaire questions (3), and repairing questionnaire slides (3).

4 DISCUSSION

Combining the overall scale data and scores data results, different display devices had almost no impact on the subjective evaluation of the experience, especially on usability, workload and overall view. This is similar to the results of related literature. HMD and large screen each had their advantages and disadvantages. HMD could provide a better gaming experience, but the problem of simulator

sickness could not be ignored. The evaluation results of different usage postures differed significantly from the related literature. There was no significant difference in simulator dizziness between sitting and standing positions. This lack of distinction was also reflected in the gaming experience of the HMD version. Although standing posture has advantages for workload, sitting posture is preferred for users in large screen mode.

The observation records of the experimental process supported the above results. We found different reactions when participants used the HMD version. 1) The more users showed excitement and joyful mood (more extensive range of movements; more verbal descriptions or interjections). 2) The faster mastery speed of usage they had (when the order of HMD-related sessions was more forward, the better rowing fluency of the following sessions; advantages in the number of gold coins obtained and distance travelled). 3) Users had more attempts to interact with the system (more looking around and on the boats, trying to get more gold coins and farther from a distance). However, two participants needed more adaptation time or breaks to relieve discomfort when completing the HMD sessions. One participant even felt slightly unwell on the day after the experiment. For most people, HMD is more popular; for sensitive people, the large screen version or the sitting position is relatively more suitable for this type of sports game.

The small sample size is a limitation of this study. As pointed out by Lewis et al., SUS validity was significantly related to sample size [10], which is more correlated than objective usability assessments (such as successful task completions (success rate) [11]. The other questionnaires also need a larger sample size to obtain universalized results, which is one of the follow-up works to extend the number of participants.

5 CONCLUSION AND FUTURE WORK

This pilot study explored VR exercise games' user preferences for using different postures (standing or sitting), display devices (VR or a large screen), and game tasks (collect coins vs distance travelled). Users did not have a significant preference for a display device (HMD/large screen) when they played VR exercise games. Similar results were also in using posture preference. However, different postures affected the user experience when they used the large screen. They preferred the sitting posture in this case. Compared with postures, display devices could change the strategies of game tasks. The HMD version could motivate users on coin collection and boating distance. This study also explored potential improvements to the experiment and VR games by providing feedback suggestions. The results contribute to discovering the opportunities and challenges to enhance such games' usability, user experience, and performance and test the experiment environment for the giant experiment and future work exploring VR game evaluation in healthcare. Moreover, we would try to study the variable outcomes associated with different VR exercise games. As an outcome of this pilot study, a user evaluation will be added for the main experiment that explores virtual interaction versus perceived realism in locomotion.

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REFERENCES

- [1] Sara Arlati, Vera Colombo, Daniele Spoladore, Luca Greci, Elisa Pedrolì, Silvia Serino, Pietro Cipresso, Karine Goulene, Marco Stramba-Badiale, Giuseppe Riva, et al. 2019. A social virtual reality-based application for the physical and cognitive training of the elderly at home. *Sensors* 19, 2 (2019), 261.
- [2] John Brooke. 1996. Sus: a 'quick and dirty' usability. *Usability evaluation in industry* 189, 3 (1996), 189–194.
- [3] Yu Fu, Yan Hu, and V Sundstedt. 2021. A Survey on AR/VR Games for Mental and Physical Public Health. *Proceedings of the eTELEMED* (2021).
- [4] Yu Fu, Yan Hu, and Veronica Sundstedt. 2022. A systematic literature review of virtual, augmented, and mixed reality game applications in healthcare. *ACM Transactions on Computing for Healthcare (HEALTH)* 3, 2 (2022), 1–27.
- [5] Yu Fu, Yan Hu, Veronica Sundstedt, and Cecilia Fagerstrom. 2021. A Survey of Possibilities and Challenges with AR/VR/MR and Gamification Usage in Healthcare. In *14th International Joint Conference on Biomedical Engineering Systems and Technologies (BIOSIGNALS)/14th Int Conf on Bio-inspired Systems and Signal Processing (BIODEVICES)*. SciTePress, 733–740.
- [6] Yu Fu, Yan Hu, Veronica Sundstedt, and Yvonne Forsell. 2022. Conceptual Design of an Extended Reality Exercise Game for the Elderly. *Applied Sciences* 12, 13 (2022), 6436.
- [7] Yan Hu, Majed Elwady, and Hans-Jürgen Zepernick. 2021. On the Effect of Standing and Seated Viewing of 360° Videos on Subjective Quality Assessment: A Pilot Study. *Computers* 10, 6 (2021), 80.
- [8] Kyle Johnsen, Sun Joo Ahn, James Moore, Scott Brown, Thomas P Robertson, Amanda Marable, and Aryabrat Basu. 2014. Mixed reality virtual pets to reduce childhood obesity. *IEEE transactions on visualization and computer graphics* 20, 4 (2014), 523–530.
- [9] Robert S Kennedy, Norman E Lane, Kevin S Berbaum, and Michael G Lillenthal. 1993. Simulator sickness questionnaire: An enhanced method for quantifying simulator sickness. *The international journal of aviation psychology* 3, 3 (1993), 203–220.
- [10] James R Lewis. 2018. The system usability scale: past, present, and future. *International Journal of Human-Computer Interaction* 34, 7 (2018), 577–590.
- [11] James R Lewis and Jeff Sauro. 2009. The factor structure of the system usability scale. In *Human Centered Design: First International Conference, HCD 2009, Held as Part of HCI International 2009, San Diego, CA, USA, July 19–24, 2009 Proceedings 1*. Springer, 94–103.
- [12] Leonard A Marascuilo and Ronald C Serlin. 1988. *Statistical methods for the social and behavioral sciences*. WH Freeman/Times Books/Henry Holt & Co.
- [13] Douglas C Montgomery. 2017. *Design and analysis of experiments*. John Wiley & sons.
- [14] Anne Mulhall. 2003. In the field: notes on observation in qualitative research. *Journal of advanced nursing* 41, 3 (2003), 306–313.
- [15] Leticia Neira-Tovar and Ivan Castilla Rodriguez. 2017. A virtual reality tool applied to improve the effects on chronic diseases-case: Emotional effects on T2DM. In *Virtual, Augmented and Mixed Reality: 9th International Conference, VAMR 2017, Held as Part of HCI International 2017, Vancouver, BC, Canada, July 9–14, 2017, Proceedings 9*. Springer, 417–425.
- [16] PERKINSOIE and XR Association. 2021. 2021 AUGMENTED AND VIRTUAL REALITY SURVEY REPORT. <https://www.perkinscoe.com/content/designinteractive/xr2021/>.
- [17] Lehana Thabane, Jinhui Ma, Rong Chu, Ji Cheng, Afisi Ismaila, Lorena P Rios, Reid Robson, Marroon Thabane, Lora Giangregorio, and Charles H Goldsmith. 2010. A tutorial on pilot studies: the what, why and how. *BMC medical research methodology* 10, 1 (2010), 1–10.
- [18] Mateus Trombetta, Patricia Paula Bazzanella Henrique, Manoela Rogofski Brum, Eliane Lucia Colussi, Ana Carolina Bertolotti De Marchi, and Rafael Rieder. 2017. Motion Rehab AVE 3D: A VR-based exergame for post-stroke rehabilitation. *Computer methods and programs in biomedicine* 151 (2017), 15–20.
- [19] Jeffrey Yim and TC Nicholas Graham. 2007. Using games to increase exercise motivation. In *Proceedings of the 2007 conference on Future Play*. 166–173.
- [20] Soojeong Yoo and Judy Kay. 2016. VRun: running-in-place virtual reality exergame. In *Proceedings of the 28th Australian Conference on Computer-Human Interaction*. 562–566.